AI-Based Hyper-Local Aviation Weather Forecast Verification System for Sri Lankan Monsoon Patterns

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Project Proposal Report

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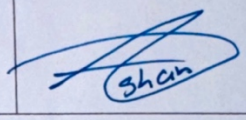
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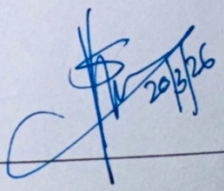
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Weather conditions such as wind speed, wind direction, temperature, and atmospheric pressure play a critical role in aviation safety and operational planning. Accurate short-term weather prediction is particularly important for aircraft takeoff and flight route planning. However, existing meteorological prediction models often provide generalized forecasts that require manual interpretation by aviation forecasters. During the Southwest Monsoon season, wind patterns can change rapidly, and the predictions generated by current models are sometimes insufficiently accurate for aviation-specific decision making. As a result, forecasters must manually analyze multiple model outputs to determine safe environmental conditions before aircraft takeoff.

This research proposes an intelligent aviation weather support system that integrates machine learning–based wind prediction with a pilot interaction platform. The system analyzes historical meteorological datasets to identify seasonal wind patterns during the Southwest Monsoon and trains predictive models capable of forecasting wind speed and direction relevant to aviation operations. A key novelty of the proposed system is the development of a runway-oriented wind prediction approach that converts predicted wind directions into headwind and crosswind components aligned with runway orientation. Additionally, a pilot portal will be developed where pilots can log in and enter flight details such as destination airport code and flight time to obtain validated weather guidance approved by meteorological forecasters.

The system is expected to improve the accuracy and usability of aviation weather predictions, enhance pilot decision-making, and support safer flight operations. From a social and commercial perspective, the proposed solution can contribute to safer aviation operations, improved forecasting efficiency, and potential integration with airport meteorological systems.

Keywords: Aviation Weather Prediction, Machine Learning, Wind Pattern Forecasting, Southwest Monsoon, Pilot Decision Support System.

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Table of Contents

DECLARATION 2
ABSTRACT 3
ACKNOWLEDGEMENT 4
TABLE OF CONTENT ..... Error! Bookmark not defined.
LIST OF TABLES 6
LIST OF FIGURES 6
LIST OF ABBREVIATIONS 7
LIST OF APPENDICES ..... Error! Bookmark not defined.
1. INTRODUCTION 8
1.1 Background and Context 8
1.2 Literature Review 9
1.3 Research Gap 11
2 OBJECTIVE 12
2.1 Main Objective 12
2.2 Specific Objectives 12
3 METHODOLOGY 13
3.1 Research Approach 13
3.2 Technical Methods and Algorithms 13
3.3 Tools and Platforms with Justification 13
3.4 Data Collection and Ethical Considerations 13
3.5 Validation Strategy and Performance Metrics 14
- Overview of the Proposed Solution and Its Components 14
4.1 System Diagram of Components 15
individual component 16
4.2 Component Interactions 17
5 USER REQUIREMENTS 18
5.1 Stakeholder Identification 18
5.2 Functional Requirements 18
5.3 Non-Functional Requirements 19
6 COMMERCIALIZATION PLAN 20
7 BUDGET AND JUSTIFICATION 21
8 WORK BREAKDOWN STRUCTURE (WBS) 22
9 GANTT CHART 23
10 REFERENCES LIST 24
11. APPENDICES 25
11.1 Appendix A: Survey Instruments 25
11.2 Appendix B: Preliminary UI Mockup 25

11.3 Appendix C: Risk Analysis 27
11.4 Appendix D– Ethical Considerations 27

LIST OF TABLES

Table 1-List of Abbreviations.....7
Table 2 project Implementation Timeline (Gantt Chart).....23
Table 3-Risk analysis.....27

LIST OF FIGURES

Figure 1 System Architecture Diagram 16
Figure 2-Individual Component Workflow Diagram 17
Figure 3-platfome user interface for piolet.....25
Figure 4-platfome user interface for piolet.....26
Figure 5-platfome user interface for piolet.....26

LIST OF ABBREVIATIONS

AI	Artificial Intelligence
ML	Machine Learning
TAF	Terminal Aerodrome Forecast
METAR	Meteorological Aerodrome Report
GFS	Global Forecast System
ECMWF	European Centre for Medium-Range Weather Forecasts
ConvLSTM	Convolutional Long Short-Term Memory
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
QNH	Atmospheric Pressure at Mean Sea Level
SDG	Sustainable Development Goal
UI	User Interface
WBS	Work Breakdown Structure

Table 1-List of Abbreviations

1. INTRODUCTION

1.1 Background and Context

Weather forecasting plays a critical role in aviation safety and operational decision-making. Pilots rely on aviation weather reports such as the Terminal Aerodrome Forecast (TAF) and Meteorological Aerodrome Report (METAR) to obtain atmospheric information before aircraft take-off and landing. These reports provide important parameters including wind speed, wind direction, temperature, atmospheric pressure, and humidity, which directly influence aircraft performance and airport operations. Accurate weather information is therefore essential for safe flight planning and runway management in modern aviation systems.

However, aviation weather forecasting is particularly challenging in tropical regions where atmospheric conditions can change rapidly. Sri Lanka experiences complex meteorological conditions influenced by seasonal monsoon circulation systems including the Southwest Monsoon, Northeast Monsoon, and two inter-monsoon periods. These seasonal systems frequently cause sudden variations in wind patterns, pressure changes, humidity levels, and convective weather conditions. Such variations can significantly affect airport operations and flight safety. Previous research has demonstrated that machine learning techniques such as artificial neural networks can successfully model monsoon rainfall patterns in Sri Lanka [1], while other studies have explored machine learning-based models for short-term weather prediction using historical meteorological datasets [2].

In aviation environments, high-impact weather hazards such as thunderstorms, lightning activity, turbulence, and strong winds are recognized as major risks to aircraft safety and operational efficiency. Studies have shown that convective weather systems can significantly affect aviation operations in tropical flight regions [3], while climatological research conducted at Bandaranaike International Airport has identified frequent lightning activity and unstable atmospheric conditions that may disrupt airport operations [4]. In addition, research on aviation meteorology highlights that severe weather phenomena remain one of the most critical challenges in flight safety management worldwide [5].

The primary stakeholders affected by these weather forecasting challenges include pilots, air traffic controllers, aviation meteorological officers, and airport operational managers who rely on accurate weather forecasts for safe flight operations. Inaccurate or delayed weather predictions can lead to flight delays, increased operational costs, and potential safety risks. Therefore, improving the accuracy and reliability of aviation weather prediction systems is essential for enhancing flight safety and operational efficiency.

Current aviation forecasting systems mainly rely on large-scale numerical weather prediction models such as the Global Forecast System (GFS) and the European Centre for Medium-Range Weather Forecasts (ECMWF) model. Although these models provide reliable global and regional predictions, they are not specifically

optimized to capture localized atmospheric variations occurring at airport terminals. As a result, there is a growing need for hyper-local aviation weather forecasting systems capable of using airport-level meteorological observations to improve short-term weather predictions.

This research addresses the identified problem by proposing an AI-based hyper-local aviation weather forecasting and verification system designed for Sri Lankan monsoon environments. The proposed system utilizes machine learning models trained on historical airport meteorological data to generate short-term forecasts and analyze seasonal weather patterns. By integrating artificial intelligence with aviation meteorology, the system aims to improve the accuracy of localized weather predictions and provide a decision-support tool for aviation stakeholders. This research belongs to the Artificial Intelligence and Data Science research cluster and contributes to global sustainability goals including SDG 9 (Industry, Innovation and Infrastructure), SDG 11 (Sustainable Cities and Communities), and SDG 13 (Climate Action) by supporting safer and more efficient aviation transportation systems.

1.2 Literature Review

Weather forecasting has been an important research area in both meteorology and aviation safety, particularly with the increasing use of artificial intelligence and machine learning techniques for atmospheric prediction. Early studies have demonstrated that machine learning approaches can effectively model complex weather patterns influenced by seasonal climatic systems. Wickramagamage [1] investigated the use of artificial neural networks to forecast monsoon rainfall patterns in Sri Lanka and showed that neural network models are capable of identifying non-linear relationships within meteorological datasets. The study highlighted the potential of machine learning algorithms to improve forecasting accuracy in regions where seasonal climate systems significantly influence weather variability.

Further research has explored machine learning techniques for short-term weather prediction using historical meteorological datasets. Jayawardena [2] proposed a machine learning-based weather prediction model for Sri Lanka that utilizes historical weather observations to forecast short-term atmospheric conditions. The study demonstrated that machine learning models can capture temporal weather patterns more effectively than traditional statistical approaches when trained using sufficient historical data. Such approaches are particularly useful in regions with rapidly changing atmospheric conditions.

In the aviation domain, weather hazards remain one of the most significant factors affecting flight safety and operational efficiency. Wang [3] analyzed aviation weather hazards within the Yangon Flight Information Region and identified phenomena such as thunderstorms, strong winds, and turbulence as critical factors affecting aviation

operations. These hazards can cause flight delays, route deviations, and increased safety risks. Similarly, Wernli and Sprenger [5] conducted a comprehensive review of high-impact weather events affecting aviation and emphasized that severe weather conditions remain a major challenge for flight safety management worldwide. Their study highlighted the importance of improving weather forecasting systems to provide reliable predictions for aviation environments.

Several studies have also examined localized atmospheric phenomena that influence airport operations. Jayasinghe [4] conducted climatological research on lightning activity at Bandaranaike International Airport and identified frequent lightning events associated with convective weather systems in the region. Such localized weather phenomena can significantly disrupt airport operations and emphasize the importance of developing forecasting models that focus on airport-level atmospheric conditions.

Recent advancements in artificial intelligence have introduced new approaches to aviation weather forecasting. Jeong [6] proposed an AI framework for automated Terminal Aerodrome Forecasting systems, demonstrating how artificial intelligence can assist meteorologists in generating aviation weather forecasts. Similarly, deep learning techniques have been widely applied to weather prediction tasks. Shi [7] introduced the Convolutional LSTM network, a deep learning architecture capable of modeling spatial and temporal relationships in atmospheric data for precipitation nowcasting. This approach has been widely recognized as an effective method for short-term weather forecasting.

Additional research has examined the broader role of deep learning in atmospheric prediction. Chen [8] reviewed various deep learning approaches used in weather forecasting and concluded that neural networks can significantly improve prediction performance when trained on large meteorological datasets. Bauer, Thorpe, and Brunet [9] discussed the challenges of integrating machine learning techniques with global weather and climate models, highlighting both the opportunities and limitations of AI-based forecasting systems. Furthermore, Rasp and Lerch [10] investigated neural network-based postprocessing techniques for ensemble weather forecasts, demonstrating that machine learning methods can improve forecast accuracy by correcting systematic prediction errors in traditional weather models.

Although these studies demonstrate the growing potential of machine learning for weather forecasting and aviation meteorology, most existing research focuses on large-scale atmospheric prediction rather than hyper-local forecasting at airport terminals. Many models are designed for global or regional weather prediction and may not adequately capture localized atmospheric variations occurring near airport environments. Additionally, limited research has addressed the influence of seasonal monsoon systems on aviation weather forecasting in tropical regions such as Sri Lanka. Therefore, there remains a need for an AI-based hyper-local aviation weather forecasting system specifically designed to analyze monsoon-driven weather patterns and improve short-term prediction accuracy at airport terminals.

1.3 Research Gap

Although modern aviation weather forecasting systems use Numerical Weather Prediction (NWP) models and observational data to generate forecasts such as METAR and TAF reports, these systems mainly provide general meteorological parameters such as wind speed and wind direction. They rarely convert these predictions into aviation-specific indicators such as headwind and crosswind relative to runway orientation. As a result, meteorological forecasters must manually interpret model outputs and determine operational conditions for pilots. Studies on airport operations indicate that wind conditions, particularly crosswinds and wind shear, significantly affect aircraft takeoff and landing safety, highlighting the need for more operationally relevant wind prediction systems. Additionally, most existing machine learning–based wind prediction research focuses on general meteorological forecasting or renewable energy applications rather than aviation decision-support systems.

Therefore, a gap exists between current meteorological prediction models and the operational needs of aviation stakeholders. Existing systems lack runway-oriented wind analysis and integrated pilot guidance platforms that can transform predicted wind parameters into actionable aviation information. To address this limitation, this research proposes a machine learning–based wind prediction system specifically designed for the Southwest Monsoon period. The system introduces a runway-oriented wind prediction approach that converts predicted wind speed and direction into headwind and crosswind components relevant to aircraft operations. In addition, a pilot portal will be developed to provide validated aviation weather guidance based on flight details, improving decision support for forecasters and enhancing aviation safety in monsoon-affected regions.

2 OBJECTIVE

2.1 Main Objective

To develop and evaluate a **machine learning–based wind pattern prediction system for the Southwest Monsoon** that converts predicted wind data into **runway-oriented headwind and crosswind components** and delivers validated aviation weather guidance to pilots through a **secure pilot portal**, achieving improved wind prediction accuracy for aviation operations.

2.2 Specific Objectives

To collect and analyze historical meteorological datasets related to wind speed and wind direction during the Southwest Monsoon period.

To identify seasonal wind patterns affecting aviation operations by analyzing historical weather data and extracting relevant features.

To develop and train machine learning models capable of predicting short-term wind speed and wind direction relevant to aviation environments.

To evaluate the performance of the developed wind prediction models using appropriate performance metrics such as RMSE, MAE, and prediction accuracy.

To design and implement a runway-oriented wind analysis module that converts predicted wind direction into headwind and crosswind components relative to runway orientation.

To develop a secure pilot portal interface that allows pilots to log in and enter flight details (destination airport code and flight time) to receive validated aviation weather guidance.

3 METHODOLOGY

3.1 Research Approach

This research follows a **Design Science Research (DSR) approach combined with experimental evaluation**. The objective is to design and develop a machine learning–based wind prediction system that addresses the real-world problem of aviation weather forecasting during the Southwest Monsoon. The study focuses on building an artifact in the form of a predictive model and a pilot interaction system. The developed models will be experimentally evaluated using historical meteorological data to assess their prediction accuracy and usefulness for aviation decision support.

3.2 Technical Methods and Algorithms

The proposed system will analyze historical meteorological data, including wind speed, wind direction, temperature, and atmospheric pressure, collected during Southwest Monsoon periods. Data preprocessing techniques such as data cleaning, normalization, and feature engineering will be applied to prepare the dataset for model training. Several machine learning algorithms, including Random Forest, Gradient Boosting, and Long Short-Term Memory (LSTM) neural networks, will be explored to predict short-term wind patterns. The predicted wind speed and direction will then be transformed into runway-oriented headwind and crosswind components using vector-based calculations based on runway orientation. These results will provide operationally meaningful wind information for aviation operations.

3.3 Tools and Platforms with Justification

The system will be developed using modern data science and web development technologies. **Python** will be used as the primary programming language due to its extensive support for data analysis and machine learning. Libraries such as **Pandas and NumPy** will be used for data processing, while **Scikit-learn and TensorFlow** will support machine learning model development. A **web-based pilot portal** will be developed using modern web technologies such as React to provide a user-friendly interface for pilots. A database system such as **MongoDB or PostgreSQL** will be used to store meteorological datasets, prediction results, and user inputs. These tools were selected because they are widely used, scalable, and well suited for predictive analytics applications.

3.4 Data Collection and Ethical Considerations

Historical meteorological data will be collected from publicly available weather databases and meteorological agencies. The dataset will include parameters such as wind speed, wind direction, temperature, and atmospheric pressure for multiple years during the Southwest Monsoon season. Since the research primarily uses environmental data rather than personal or sensitive information, ethical risks are minimal. However, all datasets will be used responsibly and properly cited to ensure compliance with data usage policies and research ethics.

3.5 Validation Strategy and Performance Metrics

To evaluate the effectiveness of the proposed prediction models, the dataset will be divided into **training and testing sets**, and validation techniques such as **cross-validation** will be applied. The performance of the models will be assessed using standard forecasting metrics such as **Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and prediction accuracy**. Additionally, the practical usefulness of the system will be evaluated by analyzing the accuracy of the calculated **headwind and crosswind components relative to runway orientation**. The results will be compared with existing meteorological predictions to determine whether the proposed system provides improved decision support for aviation operations.

4 HIGH-LEVEL SYSTEM ARCHITECTURE

- Overview of the Proposed Solution and Its Components

The proposed solution is an **aviation weather decision-support system** designed to improve wind prediction accuracy during the **Southwest Monsoon** and deliver operationally useful weather guidance for aviation stakeholders. The system combines **historical meteorological data analysis, machine learning–based wind prediction, runway-oriented wind interpretation, forecaster validation, and pilot-facing weather delivery** within one integrated platform. The architecture is designed to support the full process from data collection to final forecast delivery. The main components of the system are the **Meteorological Data Collection Module**, **Data Preprocessing and Feature Engineering Module**, **Wind Prediction Model**, **Runway-Oriented Wind Conversion Module**, **Forecaster Review and Approval Module**, **Pilot Portal**, and **Database and Storage Layer**. These components work together to transform raw weather observations into validated aviation-specific weather guidance.

4.1 System Diagram of Components

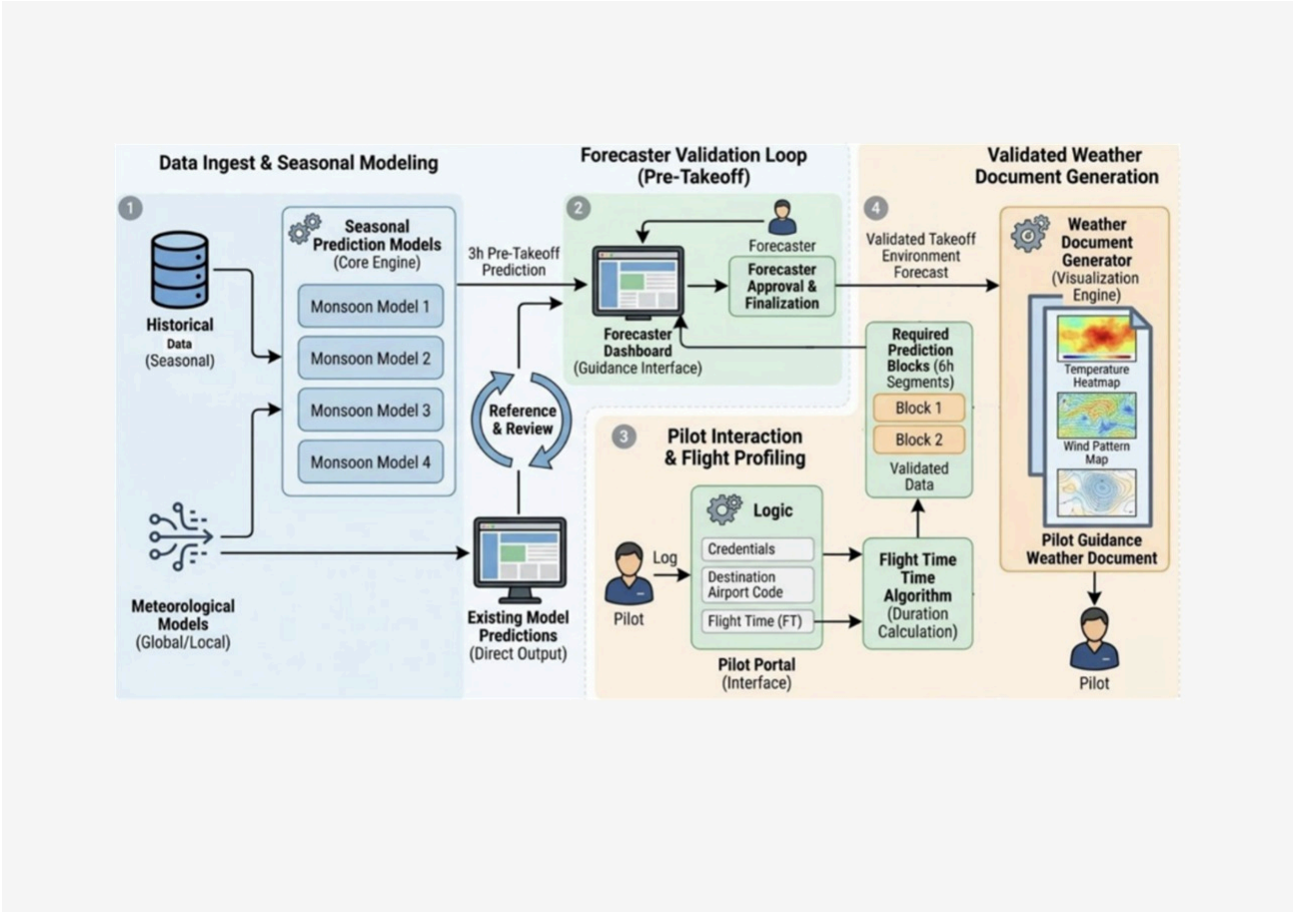


Figure 1 System Architecture Diagram

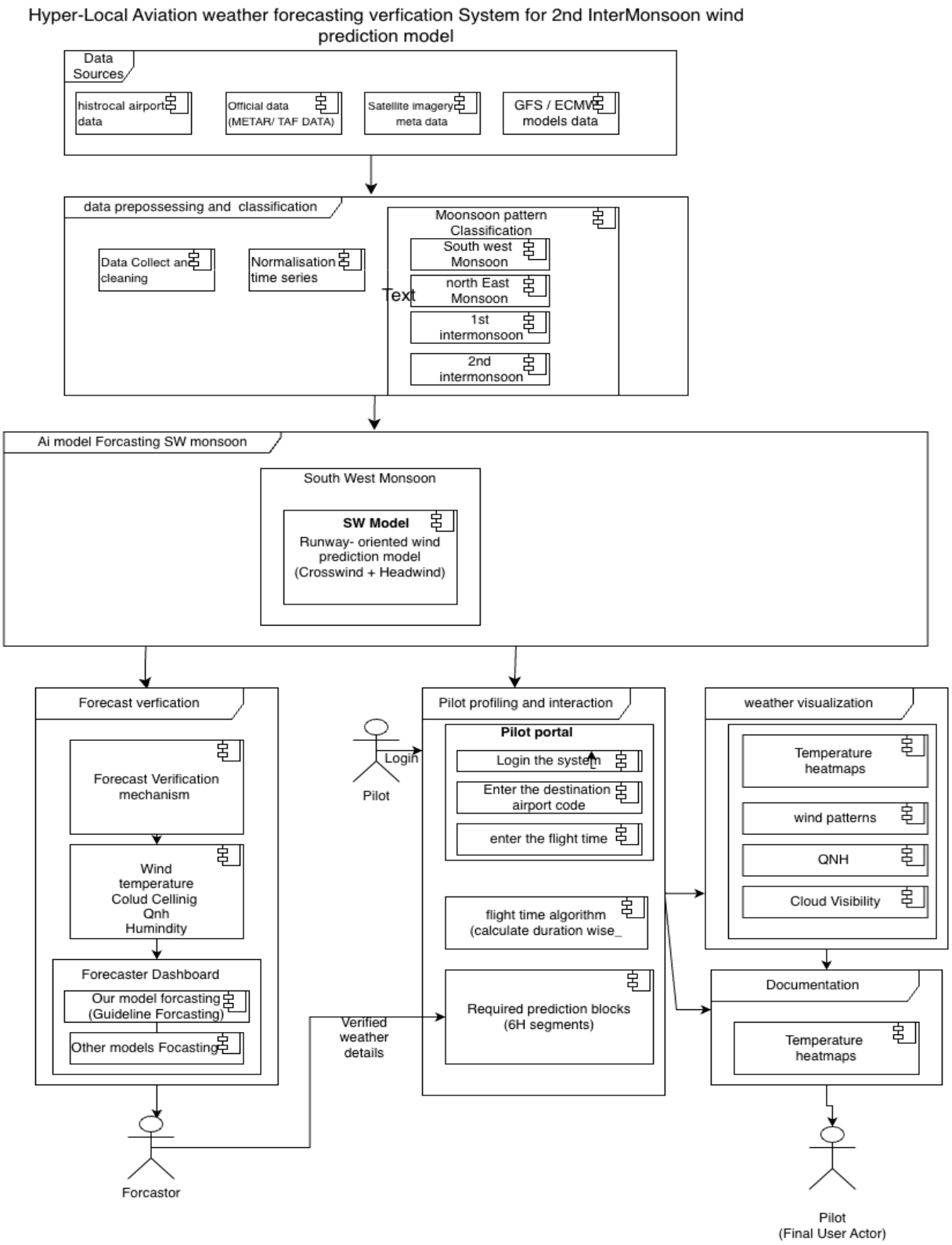


Figure 2 System Architecture Diagram

individual component

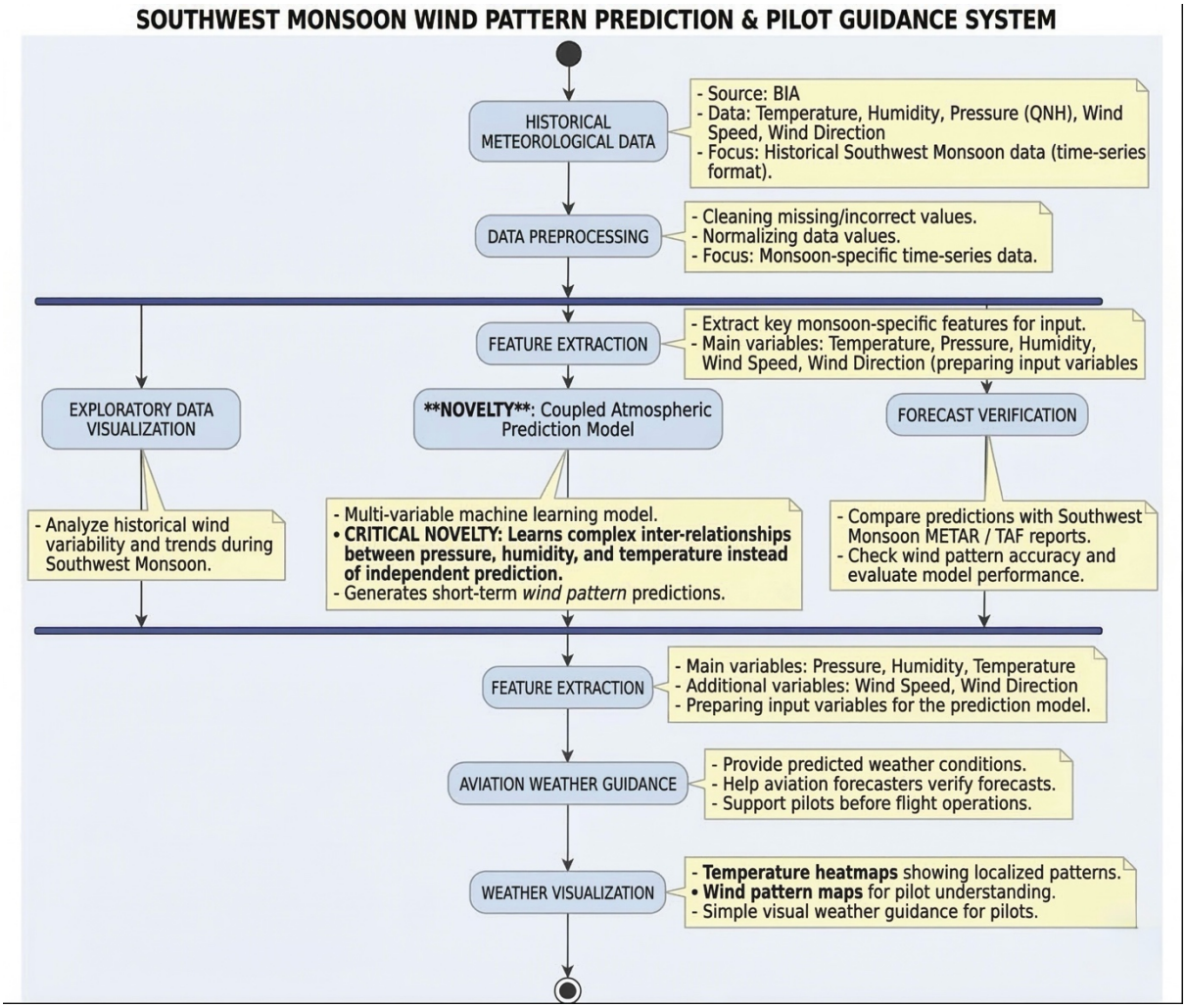


Figure 3-Individual Component Workflow Diagram

4.2 Component Interactions

The system begins by collecting data from meteorological sources, after which the preprocessing module cleans and transforms the data into a suitable format for machine learning. The trained wind prediction model then generates short-term wind forecasts using the processed inputs. These outputs are passed to the runway-oriented conversion module, which calculates crosswind and headwind values relative to runway orientation. The converted forecast is then sent to the forecaster review module, where a meteorological forecaster validates or adjusts the result before approval. Once approved, the finalized forecast is stored in the database and made available to the pilot portal. When a pilot logs in and enters flight details, the system retrieves the relevant approved forecast and displays it in an understandable format.

5 USER REQUIREMENTS

5.1 Stakeholder Identification

The proposed system involves several key stakeholders who interact with or benefit from the aviation weather prediction system. The primary stakeholders are pilots who require accurate and operationally relevant weather information before flight operations. **Meteorological forecasters** are also important stakeholders because they analyze model predictions, validate forecasts, and provide the final approved weather guidance. Airport authorities and air traffic management personnel rely on accurate wind information for runway management and operational planning. In addition, system administrators are responsible for maintaining the system, managing user access, and ensuring that the system operates securely and efficiently. These stakeholders collectively influence the requirements and functionality of the proposed system.

5.2 Functional Requirements

The functional requirements define the core operations that the system must perform to support aviation weather prediction and pilot guidance.

1. The system shall **collect and store historical meteorological data** including wind speed, wind direction, temperature, and pressure.
2. The system shall **preprocess meteorological data** by performing data cleaning, normalization, and feature extraction.
3. The system shall **train machine learning models** to predict wind speed and wind direction during the Southwest Monsoon period.
4. The system shall **convert predicted wind values into runway-oriented headwind and crosswind components** relevant to aircraft operations.
5. The system shall **allow meteorological forecasters to review and approve predicted forecasts** before they are shared with pilots.
6. The system shall **provide a pilot portal where pilots can log in using secure credentials**.
7. The system shall **allow pilots to enter flight details** such as destination airport code and flight time.
8. The system shall **generate and display validated aviation weather guidance** based on the pilot's flight details.
9. The system shall **store forecast records and prediction results** for future analysis and system improvement.
10. The system shall **allow administrators to manage users and system configurations**.

5.3 Non-Functional Requirements

Non-functional requirements describe the quality attributes and performance expectations of the system.

- **Security: The** system must implement secure authentication and authorization mechanisms to ensure that only authorized users can access weather forecasts and system features. User credentials and sensitive information must be securely stored and protected.
- **Performance: The** system should process meteorological data and generate wind predictions efficiently so that pilots and forecasters receive results within an acceptable response time.
- **Usability: The** pilot portal and forecasting interface should be user-friendly, allowing users to easily input flight details and interpret weather information without requiring advanced technical knowledge.
- **Reliability: The** system should consistently provide accurate predictions and remain operational without frequent system failures.
- **Scalability: The** architecture should support expansion to additional airports, weather models, and users without significant performance degradation.
- **Maintainability: The** system should be designed so that models, datasets, and system components can be updated or improved with minimal disruption to system operations.

6 COMMERCIALIZATION PLAN

The proposed AI-based hyper-local aviation weather forecasting and verification system is developed mainly as a research prototype to demonstrate the use of machine learning techniques in aviation meteorology. The system focuses on analyzing local atmospheric conditions and improving short-term aviation weather prediction using meteorological data from Bandaranaike International Airport.

This project is conducted for academic research purposes and is related to a government domain. Therefore, the developed system is not intended to be commercially sold or used for revenue generation. Business models such as pricing strategies or subscription-based services are not considered within the scope of this study.

Instead, the main purpose of the system is to contribute to public safety and social benefit by supporting more reliable aviation weather forecasting. Accurate weather predictions can help aviation authorities and pilots make safer operational decisions, reducing weather-related risks in flight operations. Therefore, the research aims to provide a technological contribution that supports aviation safety and benefits society

7 BUDGET AND JUSTIFICATION

- **Hardware Cost**

  Basic hardware resources such as a personal computer or laptop with sufficient processing power and memory are required for data preprocessing, machine learning model development, and system implementation. The project will mainly utilize existing computing devices; however, minor hardware upgrades such as additional storage (SSD or external drive), RAM expansion, or other small accessories may be required to improve processing performance during model training and data analysis.

  Estimated Cost: Rs. 15,000

- **Software and Cloud Services**

  The project will primarily use open-source software tools such as Python, TensorFlow, Scikit-learn, and data visualization libraries for machine learning model development and weather data analysis. In addition, cloud services such as Amazon Web Services (AWS) may be used to store meteorological datasets and perform computational experiments for model training and testing. Since most software tools are open-source, the cost mainly covers cloud service usage and minor development tools.
  Estimated Cost: RS. 20000

- **Processing Resources**

  Additional computational resources may be required to efficiently process large meteorological datasets and run machine learning algorithms. These resources will support data preprocessing, model training, and evaluation tasks required for the aviation weather forecasting system.
  Estimated Cost: Rs. 10,000

  Budget Justification

This study is conducted as an academic research project with limited financial requirements. Most of the required resources, including computing devices and development tools, are already available and open-source software will be used for machine learning model development and data visualization. Cloud services such as AWS will only be used on a limited scale for storing meteorological datasets and running computational experiments. Since this research is conducted for academic and experimental purposes and is not intended as a commercial system to be sold to government authorities, the budget allocation remains minimal. The estimated costs mainly support basic computational resources required to conduct and evaluate the proposed aviation weather forecasting system.

8 WORK BREAKDOWN STRUCTURE (WBS)

Task ID	Component	Task Description	Timeline	Key Deliverables
1	ML Development	Southwest Monsoon Wind Prediction: Develop a multi-variable machine learning model to analyze and predict wind speed and directional variations.	Jun – Aug	Trained ML Model for SW Monsoon wind patterns.
2	Novelty Integration	Coupled Atmospheric Modeling: Implement the inter-relationship logic where wind predictions are influenced by concurrent pressure and temperature shifts.	Jun – Aug	Logic module for Coupled Atmospheric Prediction.
3	Aviation Logic	Runway-Oriented System: Create an algorithm to calculate crosswind and headwind components specifically for airport runway orientations.	Jul – Aug	Runway-specific wind data output.
4	Software Module	Pilot Portal & Authentication: Develop the front-end and back-end for the pilot login system and flight detail input validation (ICAO codes/Flight Time).	Jul – Aug	Functional Pilot Portal (UI/UX) & Authentication Module.
5	Visualization	Wind Pattern Mapping: Generate dynamic visual representations (vectors/streamlines) for the Pilot Guidance Weather Document.	Aug – Sep	Visual Wind Pattern Maps for final reports.

9 GANTT CHART

Task	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Literature Review & Problem identification										
Proposal Preparation & Submission										
Data Collection (Airport Meteorological Data)										
Data Preprocessing & Feature Engineering										
Development of Coupled Prediction Model										
Model Evaluation & Forecast verification										
Weather Visualization Development										
System Testing & Result Validation										
Final Report & Website Development										
Final Presentation & Viva										
Research Paper Submission										
Research Paper Publication Evidence										

Table 2 project Implementation Timeline (Gantt Chart)

10 REFERENCES LIST

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11.APPENDICES

11.1 Appendix A: Survey Instruments

This research does not involve surveys or questionnaires because the study focuses on meteorological data analysis and machine learning–based weather prediction. Therefore, no survey instruments are required for this project.

11.2 Appendix B: Preliminary UI Mockup

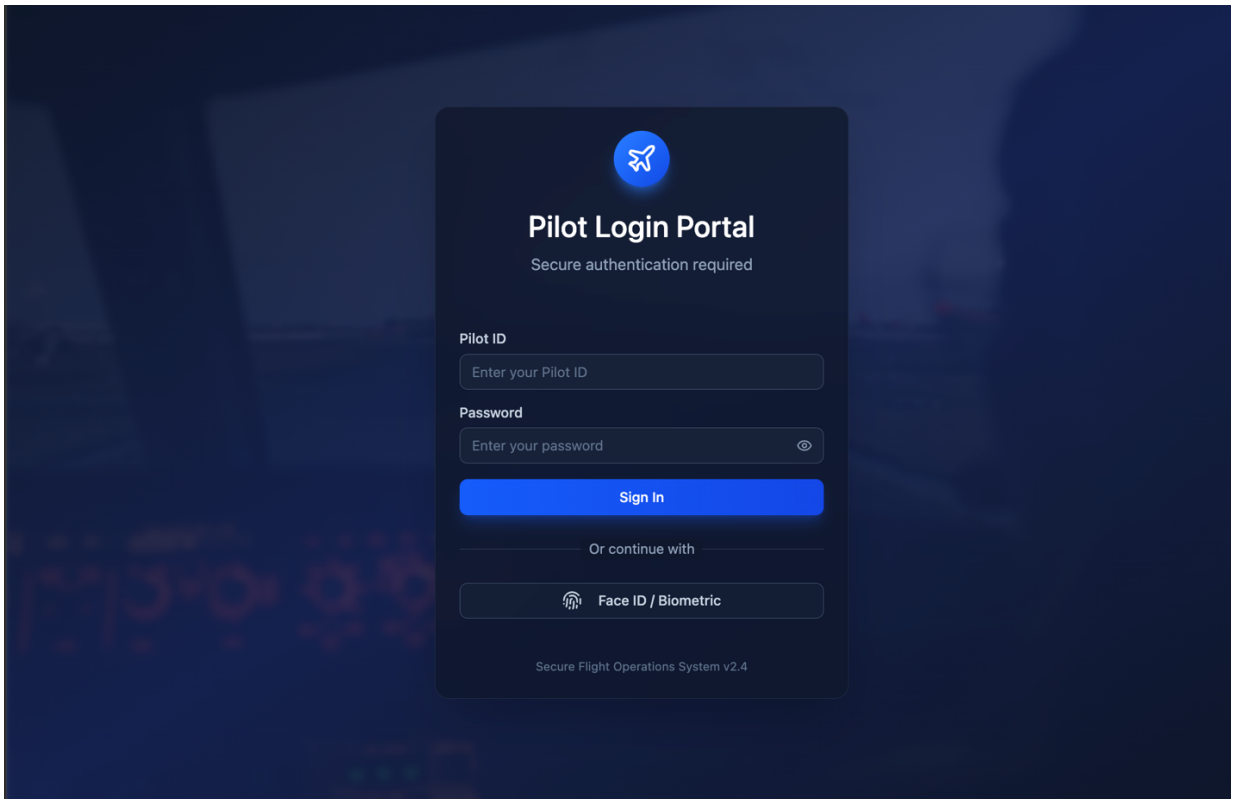


Figure 4-platfome user interface for piolet

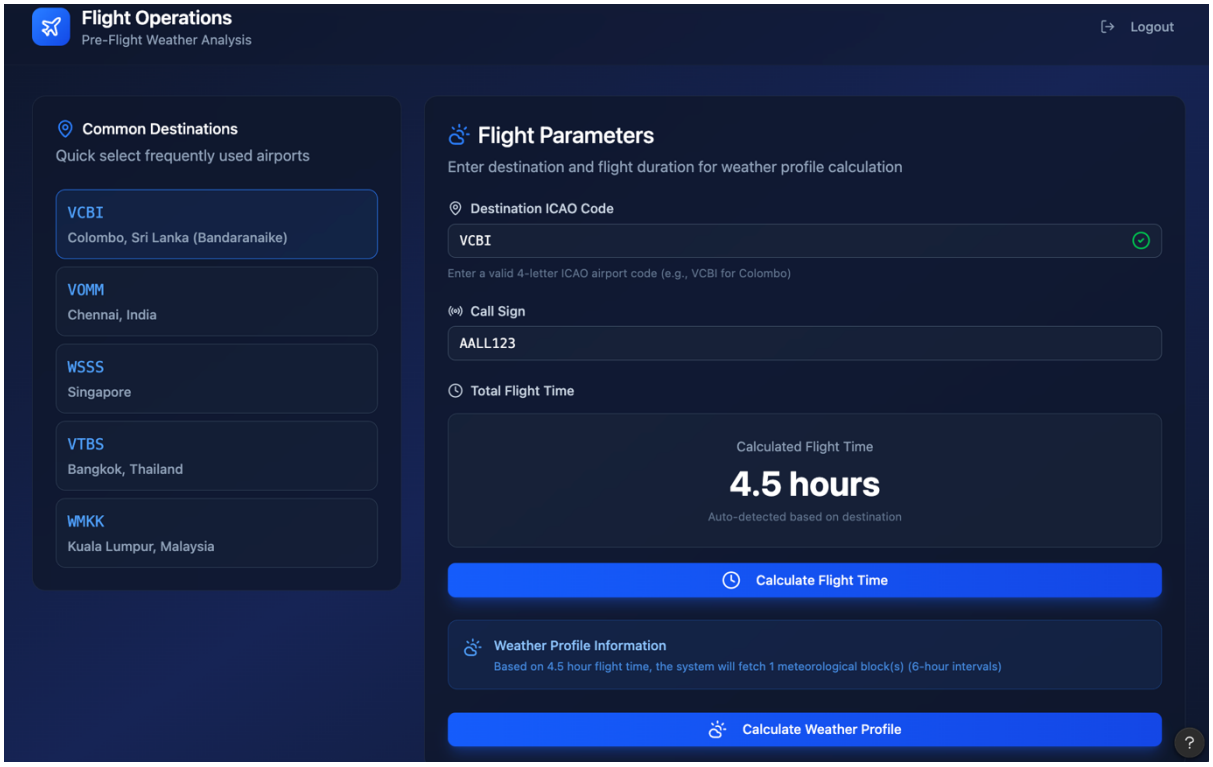


Figure 5-platfome user interface for piolet

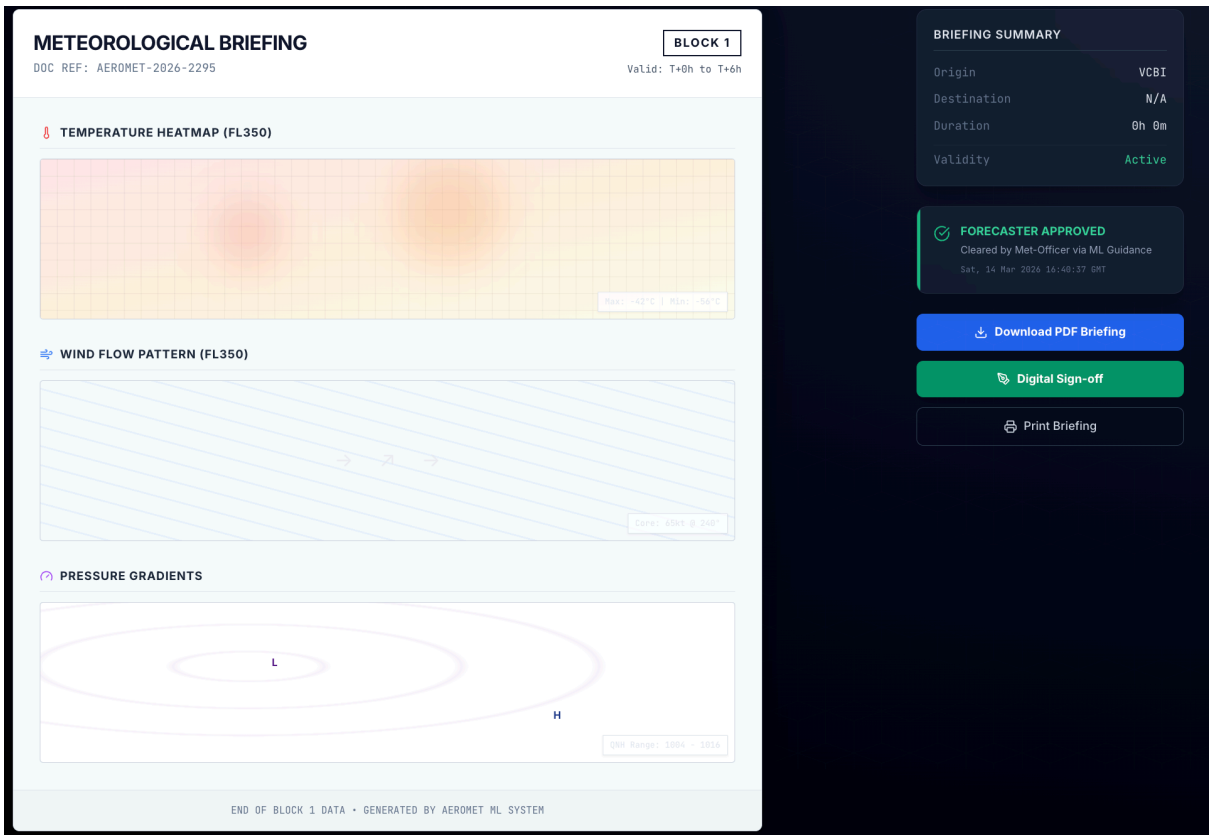


Figure 6-platfome user interface for piolet

11.3 Appendix C: Risk Analysis

The risk analysis appendix identifies potential project risks such as **data availability issues, model accuracy limitations, system integration challenges, or deployment constraints**. Mitigation strategies will also be outlined to reduce the impact of these risks.

Risk	Description	Mitigation
Data Quality Issues	Weather data may contain missing values	Apply data preprocessing and cleaning
Model Accuracy	ML predictions may not always be accurate	Use multiple models and validation
System Integration	Difficulty integrating with airport systems	Design modular architecture
Limited Dataset	Insufficient historical weather data	Use additional meteorological sources

Table 3-Risk analysis

11.4 Appendix D– Ethical Considerations

This research does not involve human subjects or personal data. The system uses meteorological data obtained through discussions and collaboration with the Meteorology Department and is supplemented with publicly available aviation weather observations.

Ethical considerations include:

- Proper acknowledgment and citation of all meteorological datasets provided by the Meteorology Department and public sources.
- Ensuring data integrity, accuracy, and responsible handling of all datasets.
- Using the system strictly for decision-support and research purposes, not for direct operational control.
- Avoiding misuse of prediction results in safety-critical aviation environments.
- Maintaining transparency and respecting the intellectual contribution of the Meteorology Department in providing the data.